



Koneru Lakshmaiah Education Foundation

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OPERATING SYSTEMS AND SYSTEMS PROGRAMMING (25CS2104E)

2026–27, Term-I

Project Problem Statement Submission Form

Section / Team Details

Section No.: 12

Team No.: 18

Project Title: Priority Scheduling

Team Members

Roll Number	Student Name	Signature
2520090102	D PAWAN KUMAR	
2520090078	HITESH SUTHAR	
2520090186	MUTYALA SAI AJITH REDDY	

2. Abstract

Priority Scheduling is a CPU scheduling project that demonstrates how an operating system selects the next process for execution according to its priority. The project models a set of processes with attributes such as process ID, arrival time, burst time, and priority, and applies priority-based scheduling to determine execution order. The solution is implemented in Linux using C/C++ and provides scheduling results such as waiting time, turnaround time, completion time, and average waiting and turnaround times. The project focuses on Operating Systems concepts including process management, CPU scheduling, ready queues, process states, scheduling policies, and process synchronization. The implementation can also demonstrate Linux process-related APIs such as `fork()`, `wait()/waitpid()`, `getpid()`, and priority-related mechanisms where required. The expected outcome is a clear Linux-based program that accepts process information, schedules processes according to priority, displays the execution sequence/Gantt-style result, and calculates performance measures for evaluating the scheduling policy.

2. Problem Statement

In a multiprogramming operating system, multiple processes may be ready to use the CPU at the same time. Selecting an appropriate process is important for reducing waiting time and improving system responsiveness. The project addresses the implementation and analysis of Priority Scheduling, in which the ready process with the highest scheduling priority is selected for execution. The project is relevant to Operating Systems because it demonstrates CPU scheduling, process management, ready-queue selection, and performance evaluation. A Linux-based C/C++ implementation will simulate the scheduling process and produce measurable scheduling results.

3. Objectives

Implement Priority Scheduling for a set of processes in a Linux environment.

Accept process details such as process ID, arrival time, burst time, and priority.

Generate the execution order and calculate completion, waiting, and turnaround times.

Analyze the scheduling performance using average waiting time and average turnaround time.

4. Proposed Methodology

The project will be developed in C/C++ and executed on Linux/Ubuntu. First, the program accepts the number of processes and their arrival time, burst time, and priority. The processes are stored in an appropriate data structure representing the ready queue. At each scheduling step, the eligible process with the highest priority is selected; if priorities are equal, a consistent tie-breaking rule such as FCFS is applied. The selected process is executed for its required burst time, after which the process is marked complete and the scheduling decision is repeated until all processes finish. The program then displays the process execution sequence and calculates completion time, turnaround time, waiting time, and their averages. Linux process APIs may be used to demonstrate process creation and synchronization where required.

5. Operating Systems Concepts / Linux APIs Used

OS Concept / Linux API / System Call	Purpose in the Project
CPU Scheduling	Selects the next ready process according to priority.
Priority Scheduling	Implements priority-based selection and execution order.
Process Management	Represents process information and tracks process states.
fork()	Creates a child process when process-creation demonstration is required.
wait()/waitpid()	Synchronizes parent and child processes and collects termination status.
getpid()	Obtains the process ID for identifying processes in Linux.

6. Individual Contribution

Roll Number	Student Name	Individual Responsibility
2520090102	D PAWAN KUMAR	Algorithm design, Priority Scheduling logic, scheduling calculations, and main C/C++ implementation.
2520090078	HITESH	Linux/process-management implementation, testing, debugging, and performance-result verification.
2520090186	AJITH REDDY	Input/output interface, result presentation, documentation, and integration testing.

7. Tools / Platforms / Software Used

Tool / Platform / Software	Purpose
Linux / Ubuntu	Development and execution environment.
C / C++	Implementation of the Priority Scheduling algorithm.
GCC / G++	Compiles the C/C++ source code on Linux.
VS Code / Text Editor	Writing, editing, and debugging source code.
Git / GitHub	Version control and project repository management.

8. Expected Outcome

The completed project will provide a Linux-based Priority Scheduling program that accepts process details, determines the priority-based execution sequence, and displays scheduling results. It will calculate completion time, turnaround time, waiting time, average waiting time, and average turnaround time. The project is expected to demonstrate practical understanding of CPU scheduling, process management, Linux system programming concepts, and performance analysis.

9. GitHub Repository

GitHub Repository Link: _____

Repository Created: _____

☐ Yes ☐ No

Faculty Name and Signature

Faculty Name: _____

Signature: _____

Remarks: _____

Faculty Use

Project Approved: _____

☐ Yes ☐ No

Date: _____

Faculty Signature: _____